Engineering Enrollment and Wages

Full-Information Estimation of the Ryoo–Rosen Model

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Code & data: `github.com/maxmaxmaxmaxmaxmax373/ryoo-rosen-mle`

Outline

1. Motivation and Contribution
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7. Conclusion

Booms and Busts in the Market for Engineers

- **Large swings:** engineering's share of all bachelor's degrees was 4.2% in 1976, 7.8% in 1985, and 4.7% in 2000.
- **Wages moved first:** engineers' median earnings relative to all college graduates rose from 1.33 to 1.50 (1972–1982).
- **Recurring debates:** talk of an engineer "shortage" in the 1950s (Arrow & Capron 1959); declining interest again around 1990.

Four Questions

1. **Supply.** How strongly does the choice of engineering respond to the wages engineers earn?
2. **Demand.** How elastic is the demand for engineers?
3. **Sources of the cycle.** What drives the booms and busts in engineering degrees: wages, or changes in the attractiveness of the field?
4. **Expectations.** Do students distinguish temporary from permanent changes in wages, or do they expect the current wage to persist?

Literature and Contribution

Literature

- **Cobweb:** Freeman (1971, 1975, 1976).
- **Rational expectations:** Siow (1984); Zarkin (1985).
- **Ryoo & Rosen (2004):** equation by equation, 1950–1990; elasticities 1.2–2.2 (demand), 2.5–4.5 (supply).
- **FIML method:** Hansen & Sargent (1980, 2013); Sargent (1989).

This Paper

1. Equilibrium solves a linear-quadratic planning problem.
2. Full-information ML with the Kalman filter, CPS & NSF data, 1968–2010.
3. Career value is a function of the state, not measured or instrumented.
4. Measurement error, cyclical AR(2) shocks, profile-likelihood intervals.
5. Rational vs. cobweb expectations in one likelihood.

Cost of joint estimation: misspecification in any one equation affects all estimates.

Our Answers

1. **Supply: highly sensitive to persistent wages.**
   A permanent 1% rise in the relative wage eventually raises the entry share by $\approx 6\%$.
2. **Demand: elasticity $\approx$ 1.6,** in line with Ryoo & Rosen; the data bound it only from above.
3. **Sources of the cycle: preferences, not wages.**
   97% of degree-share variance comes from the attractiveness of engineering, a damped cycle of $\approx$ 22 years.
4. **Expectations: the data cannot tell.**
   A cobweb model fits about as well and better matches the comovement of wages and future degrees, but only with a nearly flat demand curve.

The answer to Q4 shapes the answer to Q2: demand elasticity 1.6 (RE) vs. 6.0 (cobweb).

Model: Setup and Shocks

- Annual periods. All variables are **log ratios** of engineers to all college graduates, in deviations from trend:
  - w_t: relative wage; N_t: relative stock; n_t: enrollment share.
  - Training takes four years: degree share $\pi_t = n_{t-4}$.
- Representative firm and household, both wage takers.
- Two shocks, each an **unrestricted AR(2)**:

$$\theta_{t+1} = a_1^\theta \theta_t + a_2^\theta \theta_{t-1} + \sigma_\theta \varepsilon_{t+1}^\theta \quad \text{(demand)}$$

$$\psi_{t+1} = a_1^\psi \psi_t + a_2^\psi \psi_{t-1} + \sigma_\psi \varepsilon_{t+1}^\psi \quad \text{(preference)}$$

- If $a_1^2 + 4a_2 < 0$: complex roots $re^{\pm i\omega} \Rightarrow$ damped cycle of period $2\pi/\omega$, decay rate $r = \sqrt{-a_2}$.
- Discount factor fixed at $\beta = 0.96$.

Firms and the Stock of Engineers

The Firm

$$\max_{\{N_t^d\}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[(f_1 + \theta_t) N_t^d - \tfrac{f_2}{2} (N_t^d)^2 - w_t N_t^d \right] \implies \boxed{w_t = -f_2 N_t^d + \theta_t}$$

Static problem; in logs, $1/f_2$ is the elasticity of demand.

The Stock

$L_{t+1} = \delta L_t + S_{t-3}$; log-linearize around balanced growth g and difference out all graduates:

$$\boxed{N_{t+1} = \lambda N_t + (1 - \lambda)\, n_{t-3}}, \qquad \lambda = \frac{\delta}{1+g}.$$

λ = share of incumbents in next year's stock; each cohort is small, so the stock adjusts slowly.

Households: Enrollment and Career Value

The Household

$$\max_{\{n_t\}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[w_t N_t^h - \tfrac{d}{2} n_t^2 + d\,\psi_t n_t \right] \quad \text{s.t.} \quad N_{t+1}^h = \lambda N_t^h + (1-\lambda) n_{t-3}$$

$$\Longrightarrow \quad \boxed{n_t = \frac{1-\lambda}{d} V_t + \psi_t}, \qquad V_t = \beta^4 \, \mathbb{E}_t \sum_{j=0}^{\infty} (\beta\lambda)^j w_{t+4+j}.$$

- V_t = **career value**: expected discounted wages after graduation, weighted by the cohort's declining share in the stock.
- Long-run entry elasticity: $\eta^{LR} = \dfrac{(1-\lambda)\beta^4}{d\,(1-\beta\lambda)}$.
- **Expectations:** baseline is rational. Cobweb alternative: $\mathbb{E}_t w_{t+j} = w_t \Rightarrow n_t = \eta\, w_t + \psi_t$.

Equilibrium as a Planning Problem

Proposition 1

An allocation $\{N_t, n_t\}$ is a competitive equilibrium allocation **if and only if** it solves

$$\max_{\{n_t\}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\theta_t N_t - \tfrac{f_2}{2} N_t^2 - \tfrac{d}{2} n_t^2 + d\, \psi_t n_t \right]$$

subject to the shocks and the stock equation, with $w_t = \theta_t - f_2 N_t$. The equilibrium is unique.

- Removes the fixed point: enrollment $\leftrightarrow$ expected wages.
- State $X_t = (N_t, n_{t-1}, \ldots, n_{t-4}, \theta_t, \psi_t, \theta_{t-1}, \psi_{t-1})'$, control $u_t = n_t$.
- Riccati equation $\Rightarrow u_t = -FX_t$, $X_{t+1} = MX_t + C\varepsilon_{t+1}$, $M = A - BF$.
- Career value is linear in the current state:

$$V_t = \beta^4 G_w M^4 (I - \beta\lambda M)^{-1} X_t.$$

Data: United States, 1968–2010

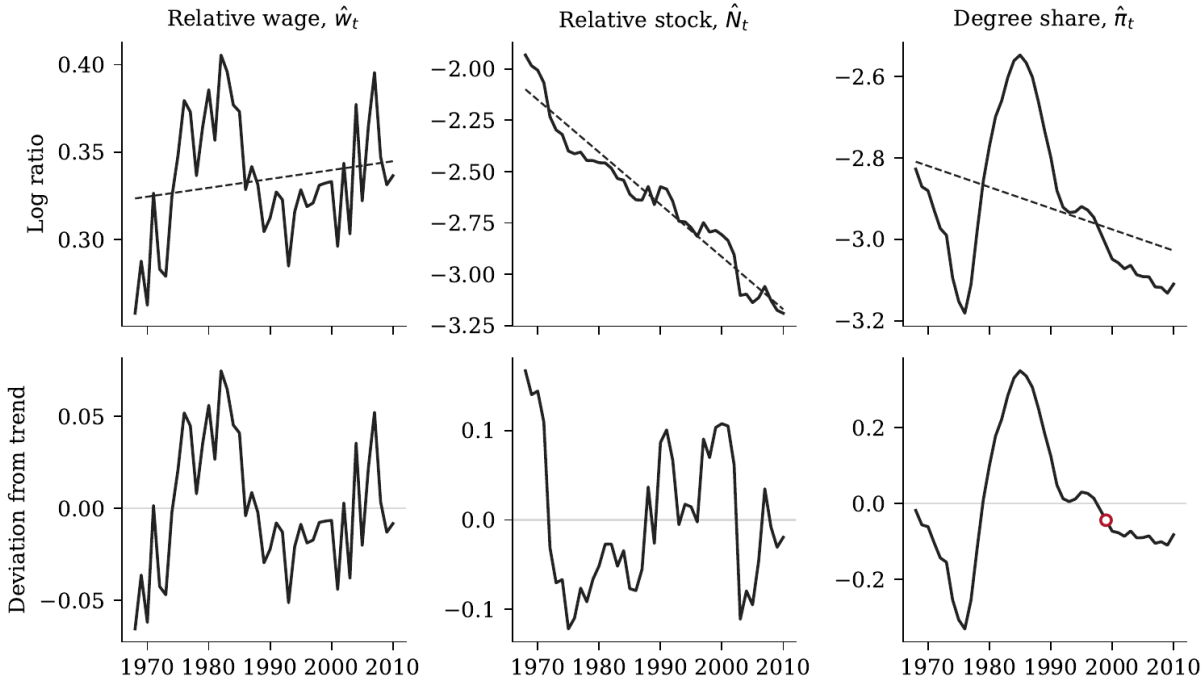


Figure 1: Relative wage, stock, and degree share: log ratios (top), detrended (bottom)

- **CPS ASEC (IPUMS):** median earnings and counts of engineers (OCC1990 044–059) vs. college graduates. **NSF:** bachelor's degrees by field.
- 43 observations; a **separate linear trend** removed from each series. Degree share: −33 log points (1976) to +35 (1985).

Estimation: Kalman-Filter Maximum Likelihood

- **Observation equation** with serially correlated measurement error:

$$y_t = GX_t + v_t, \qquad v^x_{t+1} = D_x v^x_t + \sigma_x e^x_{t+1}, \quad x \in \{w, N, \pi\},\ D_\pi = 0.$$

- **Why:** only 611–1,409 engineers per CPS year; stock falls 17.7 log points in 2003 (new occupation codes); half the sample overlaps across years.
- State-space system $\Rightarrow$ Kalman filter $\Rightarrow$ exact Gaussian likelihood:

$$\log L = \sum_{t=1}^{T} \Big[-\tfrac{3}{2}\log 2\pi - \tfrac{1}{2}\log\det\Omega_t - \tfrac{1}{2} a_t' \Omega_t^{-1} a_t \Big].$$

- 14 parameters; Nelder–Mead $\to$ BFGS $\to$ Nelder–Mead; 7 of 8 starts agree.
- Likelihood is asymmetric $\Rightarrow$ **profile-likelihood** 95% intervals: $LR(x) \le 3.84$.

Estimates: Demand, Enrollment, and Stock

Q1 (supply) and Q2 (demand)

Parameter	Description	Estimate	S.E.	Profile 95% interval
f_2	Slope of inverse demand	0.638	(0.487)	[0.135, >6.32]
d	Enrollment cost slope	0.075	(0.054)	[<0.009, 0.439]
λ	Incumbents' share	0.955	(0.029)	[0.895, >0.995]
$1/f_2$	Demand elasticity	1.57	(1.20)	
$(1-\lambda)/d$	Coefficient on V_t	0.59	(0.38)	
η^{LR}	Long-run entry elasticity	6.07	(3.63)	

- **Demand:** 1.57 lies within Ryoo–Rosen's 1.2–2.2; data rule out a nearly flat curve (elasticity > 7.4), not a steep one.
- **Enrollment:** 6.07, above RR's 2.5–4.5, imprecise; data rule out *insensitive* enrollment.
- **Stock:** $\lambda = 0.955$ vs. accounting benchmark 0.959.

Shocks, Measurement Error, and Profile Likelihood

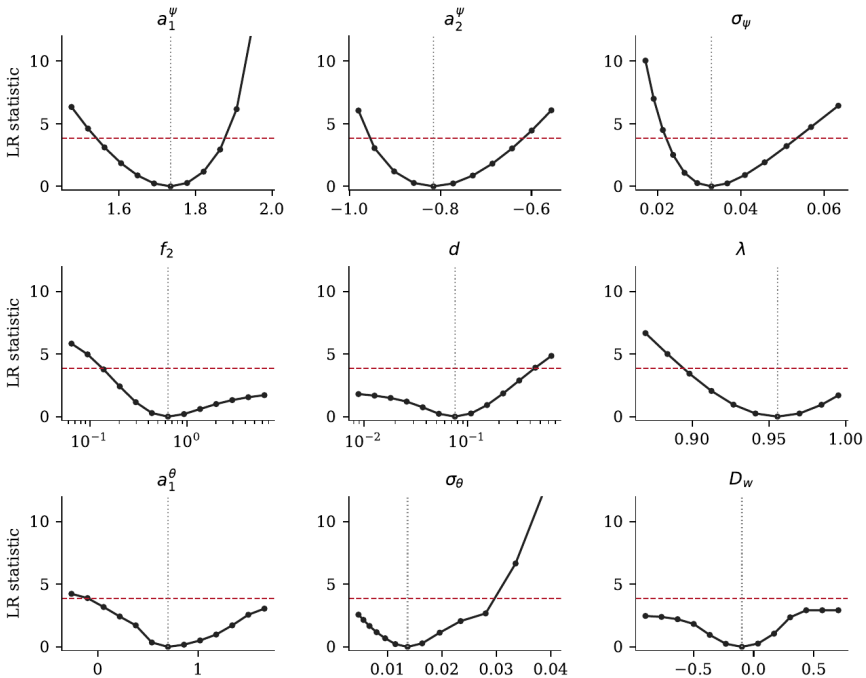


Figure 2: Profile likelihood $LR(x)$

Preference Shock ψ

Roots $0.867 \pm 0.251i$: modulus 0.90, period $\approx$ 22 years. Real roots rejected: $LR = 4.72 > 3.84$.

Demand Shock θ

Real roots 0.89, -0.20; poorly determined.

Measurement Error

Stock error s.d. 0.049, persistence 0.73, vs. 0.079 for the stock itself: a noisy series. Degree-share error at zero.

Model Fit: Data vs. Simulated Moments

		Model (1,000 samples)		
	Data	Median	5%	95%
Standard deviation				
$\tilde{w}_t$	0.035	0.028	0.021	0.040
$\tilde{N}_t$	0.079	0.064	0.046	0.088
$\tilde{\pi}_t$	0.169	0.118	0.065	0.195
First-order autocorrelation				
$\tilde{w}_t$	0.569	0.431	0.016	0.725
$\tilde{N}_t$	0.765	0.657	0.408	0.816
$\tilde{\pi}_t$	0.957	0.934	0.869	0.962
Wage and future degree share				
$\mathrm{corr}(\tilde{w}_t, \tilde{\pi}_{t+4})$	0.750	0.352	−0.118	0.691
$\mathrm{corr}(\tilde{w}_t, \tilde{\pi}_{t+5})$	0.763	0.365	−0.072	0.711

- Volatility and persistence: every data moment lies in the 5%–95% band.
- **The gap:** the wage–future degree correlation lies above the 95th percentile.

Impulse Responses (Q3)

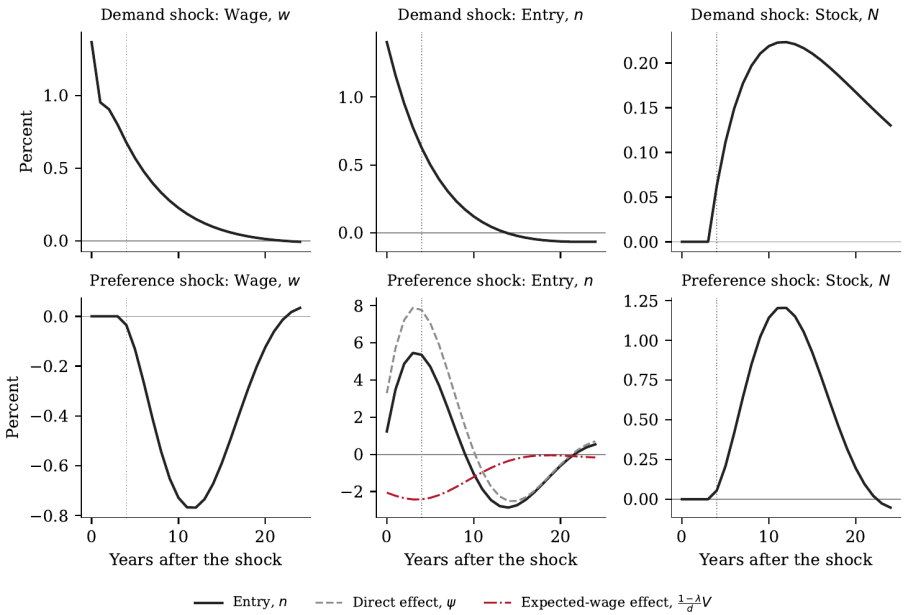


Figure 3: Demand (top) and preference (bottom) shocks

Demand shock (RE): students discount a temporary rise.

- Wage +1.37%, enrollment +1.40% at once.
- Wage gain halves in ≈ 4 years: V_t rises 2.4 pts, not 14.0.
- Response $\approx$ 1/6 of a permanent rise.

Preference shock: largely self-offsetting.

- Enrollment rises 37% of the shock.
- Foreseen glut lowers V_t, offsetting 63%.
- Stock half-life: 15.2 years.

Variance Decomposition

Q3: where the fluctuations come from

	Observed series			Model variables		
	$\tilde{w}_t$	$\tilde{N}_t$	$\tilde{\pi}_t$	w_t	N_t	n_t
Demand shock	0.42	0.01	0.03	0.56	0.07	0.03
Preference shock	0.34	0.18	0.97	0.44	0.93	0.97
Measurement error	0.24	0.81	0.00			

1. **97%** of the variance of enrollment and the degree share comes from the preference shock: swings in the choice of major reflect the attractiveness of engineering, not wages.
2. **81%** of the variance of the observed stock is measurement error.
3. Both shocks matter for wages: demand 56%, preference 44% of the true wage's variance.

Rational Expectations versus the Cobweb

Q4: do students distinguish temporary from permanent wages?

	Rational expectations	Cobweb
$\log L$	260.47	262.00
λ	0.955	0.946
Entry elasticity (η^{LR} / η)	6.07	5.10
Demand elasticity $1/f_2$	1.57	5.99
Demand shock: modulus, period	0.89, real roots	0.97, 25.5 yr
Demand innovation s.d. σ_θ	0.0137	0.0019
Preference shock: modulus, period	0.90, 22.3 yr	0.86, 16.4 yr

- Same data and likelihood; 14 parameters each, not nested: a 1.5 gap in $\log L$ **does not discriminate**.
- Agreement on λ, stock noise, and a **strong entry response**.
- Differences on the **demand side**: a nearly flat curve and an almost deterministic long demand cycle.

The Cobweb Matches the Wage–Degree Link

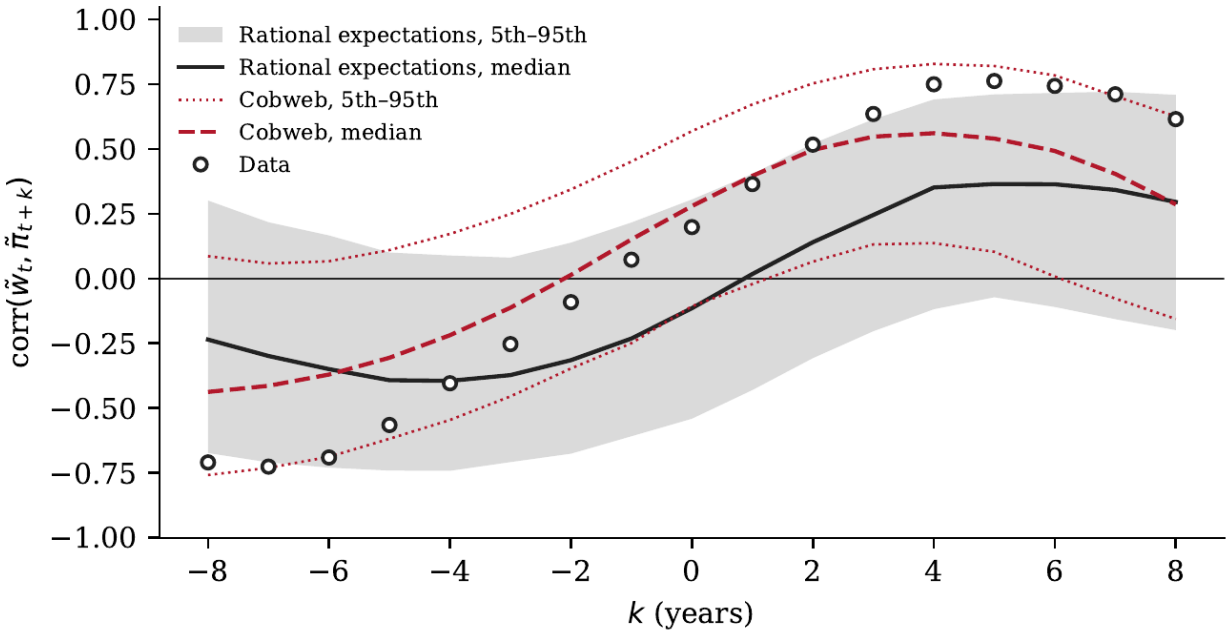


Figure 4: Correlation of the wage with the degree share k years later

- At $k = 4$: data 0.75; RE median 0.35, band $[-0.12, 0.69]$; cobweb median 0.56, band $[0.14, 0.83]$ **covers the data**.
- Cobweb students treat today's wage as permanent: a 1% wage rise raises enrollment 5.1% on impact vs. $\approx$ 1.0% under RE.

Responses under the Two Forms of Expectations

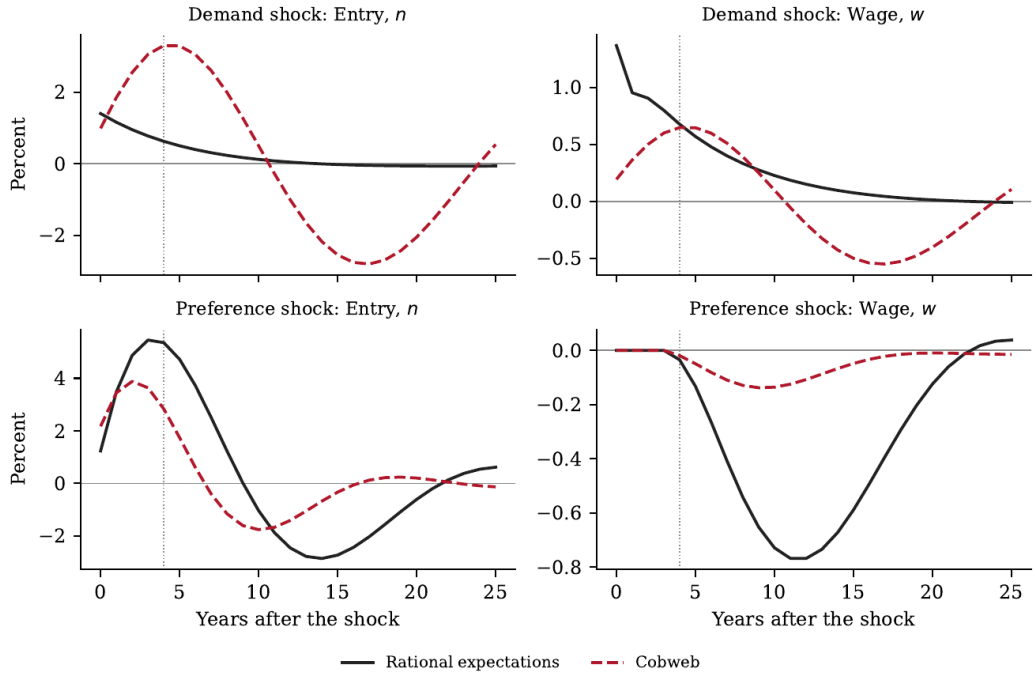


Figure 5: Solid: rational expectations; dashed: cobweb

- **Demand shock:** the classic cobweb cycle — high wage, entry surge, glut four years later, wage falls, entry below normal (partly from the demand cycle itself).
- **Preference shock:** cobweb enrollment moves one for one for four years; under RE it rises only 37% on impact.

Conclusion

Answers to the Four Questions

1. **Supply:** enrollment responds **strongly to persistent** wage changes ($\eta^{LR} \approx 6$).
2. **Demand:** elasticity ≈ 1.6 under RE, but it depends on expectations.
3. **Cycle:** $\approx 97\%$ of degree swings come from the attractiveness of engineering.
4. **Expectations:** the data cannot tell; RE vs. cobweb implies demand elasticity 1.6 vs. 6.0.

- **Limitations:** β fixed at 0.96; linear detrending; independent shocks; no enrollment adjustment costs; only 43 observations; 81% of stock variance is measurement error.
- **Extensions:** physicians, lawyers, teachers, scientists — same planning problem, state space, and likelihood.

Thank you!

Appendix A.1: Stock Accumulation

- Divide $L_{t+1} = \delta L_t + S_{t-3}$ by L_t. On a balanced growth path, $L_{t+1}/L_t = 1 + g$ and $S_{t-3}/L_t = \bar{s}$, so

$$1 + g = \delta + \bar{s}.$$

- Let $\hat{\ell}_t, \hat{s}_t$ be log deviations of L_t, S_t from the path. A first-order expansion of

$$(1+g)\, e^{\hat{\ell}_{t+1} - \hat{\ell}_t} = \delta + \bar{s}\, e^{\hat{s}_{t-3} - \hat{\ell}_t}$$

gives

$$(1+g)(\hat{\ell}_{t+1} - \hat{\ell}_t) = \bar{s}\,(\hat{s}_{t-3} - \hat{\ell}_t) \implies \hat{\ell}_{t+1} = \lambda \hat{\ell}_t + (1-\lambda)\hat{s}_{t-3},$$

with $\lambda = \delta/(1+g)$ and $1 - \lambda = \bar{s}/(1+g)$.

- Subtract the same equation for all college graduates (superscript c), with $N_t = \hat{\ell}_t - \hat{\ell}^c_t$, $n_t = \hat{s}_t - \hat{s}^c_t$:

$$N_{t+1} = \lambda N_t + (1-\lambda) n_{t-3} + \underbrace{(\lambda - \lambda^c)\left(\hat{\ell}^c_t - \hat{s}^c_{t-3}\right)}_{\text{dropped}}.$$

- In the data the last term has detrended s.d. 0.0022, against 0.0787 for N_t.

Appendix A.2: Household First-Order Condition

- Iterating the constraint $N^h_{t+1} = \lambda N^h_t + (1-\lambda) n_{t-3}$:

$$\frac{\partial N^h_s}{\partial n_t} = (1-\lambda)\lambda^{s-t-4} \quad (s \geq t+4), \qquad 0 \text{ otherwise.}$$

- Taking wages as given, the derivative of the objective w.r.t. n_t, divided by β^t:

$$-d\,(n_t - \psi_t) + (1-\lambda)\,\mathbb{E}_t \sum_{j=0}^{\infty} \beta^{4+j}\lambda^j w_{t+4+j} = 0, \qquad \text{second derivative } -d < 0.$$

  Rearranging: $n_t = \frac{1-\lambda}{d} V_t + \psi_t$ with $V_t = \beta^4 \mathbb{E}_t \sum_j (\beta\lambda)^j w_{t+4+j}$.

- **Long-run elasticity.** If w_{t+k} rises by one unit for all $k \geq 0$:

$$\Delta V_t = \beta^4 \sum_{j\geq 0} (\beta\lambda)^j = \frac{\beta^4}{1-\beta\lambda} = 10.26, \qquad \Delta n_t = \eta^{LR} = \tfrac{1-\lambda}{d} \times 10.26 \approx 6.07.$$

- **Cobweb.** $\mathbb{E}_t w_{t+j} = w_t \Rightarrow V_t = \frac{\beta^4 w_t}{1-\beta\lambda} \Rightarrow n_t = \eta w_t + \psi_t$. Using $w_t = -f_2 N_t + \theta_t$:

$$n_t = -\eta f_2 N_t + \eta\theta_t + \psi_t \equiv KX_t, \quad X_{t+1} = (A + BK)X_t + C\varepsilon_{t+1}.$$

Appendix A.3: AR(2) Processes with Complex Roots

- The impulse response $x_h = \partial x_{t+h}/\partial \varepsilon_t$ of $x_t = a_1 x_{t-1} + a_2 x_{t-2} + \varepsilon_t$ solves

$$x_h = a_1 x_{h-1} + a_2 x_{h-2}, \qquad x_0 = 1,\ x_{-1} = 0.$$

- With distinct roots $z_{1,2}$ of $z^2 = a_1 z + a_2$: $\quad x_h = \dfrac{z_1^{h+1} - z_2^{h+1}}{z_1 - z_2}$.
- If $a_1^2 + 4a_2 < 0$: $z_{1,2} = re^{\pm i\omega}$ with $r^2 = z_1 z_2 = -a_2$ and $r\cos\omega = a_1/2$, so

$$x_h = \frac{2i\, r^{h+1} \sin\big((h+1)\omega\big)}{2i\, r \sin\omega} = r^h \frac{\sin\big((h+1)\omega\big)}{\sin\omega},$$

  an oscillation with period $2\pi/\omega$ whose amplitude decays at rate r.

Application to the Preference Shock

$a_1^{\psi} = 1.735$, $a_2^{\psi} = -0.815$: $r = \sqrt{0.815} = 0.90$, $\cos\omega = 1.735/(2 \times 0.90) \Rightarrow$ period $2\pi/\omega \approx 22.3$ years.

Test of real roots, $H_0 : (a_1^{\psi})^2 + 4a_2^{\psi} \geq 0$: $LR = 4.72$. At the least favorable boundary point the statistic is $\frac{1}{2}\chi_0^2 + \frac{1}{2}\chi_1^2$ (5% critical value 2.71) $\Rightarrow$ rejected.

Appendix A.4: Proof of Proposition 1

- Since $\partial\left(\theta_s N_s - \frac{f_2}{2} N_s^2\right)/\partial N_s = \theta_s - f_2 N_s$, the planner's first-order condition for n_t is

$$d\left(n_t - \psi_t\right) = (1-\lambda)\,\mathbb{E}_t \sum_{j=0}^{\infty} \beta^{4+j} \lambda^j \left(\theta_{t+4+j} - f_2 N_{t+4+j}\right). \qquad (*)$$

- For $f_2, d > 0$ the objective is strictly concave and the constraints linear; for processes with bounded second moments, $\beta < 1$ ensures transversality. Hence $\{N_t, n_t\}$ solves the planning problem **iff** $(*)$ holds for all t, and the solution is unique.
- **Necessity.** In an equilibrium, substitute the firm's condition $w_s = \theta_s - f_2 N_s$ into the household's condition $\Rightarrow (*)$.
- **Sufficiency.** Let $\{N_t, n_t\}$ satisfy $(*)$; set $w_s = \theta_s - f_2 N_s$ and $N_s^d = N_s^h = N_s$. Then the demand curve and market clearing hold by construction, and $(*)$ becomes the household's condition. Both private problems are strictly concave, so these first-order conditions are sufficient: the allocation with this wage is an equilibrium.
- **Uniqueness** follows from that of the planner's solution. □

Appendix A.5: Linear-Quadratic Representation

$X_t = (N_t,\ n_{t-1},\ n_{t-2},\ n_{t-3},\ n_{t-4},\ \theta_t,\ \psi_t,\ \theta_{t-1},\ \psi_{t-1})'$, $\quad u_t = n_t$,
$X_{t+1} = AX_t + Bu_t + C\varepsilon_{t+1}$:

$$A = \begin{pmatrix} \lambda & 0 & 0 & 1-\lambda & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & a_1^\theta & 0 & a_2^\theta & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & a_1^\psi & 0 & a_2^\psi \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad C = \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ \sigma_\theta & 0 \\ 0 & \sigma_\psi \\ 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

- Period return $r(X_t, u_t) = X_t' R X_t + Q u_t^2 + 2 u_t W X_t$, where the only nonzero elements of R are $R_{11} = -f_2/2$ and $R_{16} = R_{61} = 1/2$; $Q = -d/2$; and W has $d/2$ in the position of ψ_t, zeros elsewhere.
- These reproduce $\theta_t N_t - \frac{f_2}{2} N_t^2 - \frac{d}{2} n_t^2 + d\, \psi_t n_t$.

Appendix A.6: Solving the Planning Problem

- Quadratic value function $v(X) = X'PX + c$; Bellman equation:

$$X'PX + c = \max_u \Big\{ X'RX + Qu^2 + 2uWX + \beta(AX + Bu)'P(AX + Bu) + \beta \operatorname{tr}(C'PC) + \beta c \Big\}.$$

- First-order condition $\Rightarrow$ decision rule

$$u_t = -FX_t, \qquad F = (Q + \beta B'PB)^{-1}(W + \beta B'PA).$$

- Matching quadratic terms $\Rightarrow$ algebraic Riccati equation

$$P = R + \beta A'PA - (W' + \beta A'PB)(Q + \beta B'PB)^{-1}(W + \beta B'PA).$$

- **Second-order condition:** B selects only n_{t-1}, so $Q + \beta B'PB \leq -d/2 < 0$. F does not depend on C: certainty equivalence. Solved after rescaling (A, B) by $\sqrt{\beta}$.
- Law of motion $X_{t+1} = MX_t + C\varepsilon_{t+1}$, $M = A - BF$; since $\mathbb{E}_t X_{t+k} = M^k X_t$,

$$V_t = \beta^4 \sum_{j \geq 0} (\beta\lambda)^j G_w M^{4+j} X_t = \beta^4 G_w M^4 (I - \beta\lambda M)^{-1} X_t.$$

Appendix B.1: State-Space System and Kalman Filter

- $s_t = (X_t', v_t')'$, $\epsilon_t = (\varepsilon_t^\theta, \varepsilon_t^\psi, e_t^w, e_t^N, e_t^\pi)' \sim$ i.i.d. $N(0, I_5)$:

$$s_{t+1} = \mathcal{M} s_t + \mathcal{C}\epsilon_{t+1}, \qquad y_t = \mathcal{G} s_t,$$

$$\mathcal{M} = \begin{pmatrix} M & 0 \\ 0 & D \end{pmatrix}, \ \mathcal{C} = \begin{pmatrix} C & 0 \\ 0 & \Sigma_v \end{pmatrix}, \ \mathcal{G} = \begin{pmatrix} G & I_3 \end{pmatrix},$$

  $D = \mathrm{diag}(D_w, D_N, 0)$, $\Sigma_v = \mathrm{diag}(\sigma_w, \sigma_N, \sigma_\pi)$; rows of G select $w_t = -f_2 N_t + \theta_t$, N_t, $\pi_t = n_{t-4}$.

- Let $\hat{s}_t = E[s_t \mid y^{t-1}]$, $\Sigma_t = \mathrm{Var}(s_t \mid y^{t-1})$, started at the stationary distribution $\hat{s}_1 = 0$, $\Sigma_1 = \mathcal{M}\Sigma_1\mathcal{M}' + \mathcal{C}\mathcal{C}'$.

- Joint normality of $(s_{t+1}, y_t) \mid y^{t-1}$ gives the recursions:

$$a_t = y_t - \mathcal{G}\hat{s}_t, \qquad \Omega_t = \mathcal{G}\Sigma_t\mathcal{G}',$$

$$K_t = \mathcal{M}\Sigma_t\mathcal{G}'\Omega_t^{-1}, \qquad \hat{s}_{t+1} = \mathcal{M}\hat{s}_t + K_t a_t,$$

$$\Sigma_{t+1} = (\mathcal{M} - K_t\mathcal{G})\Sigma_t(\mathcal{M} - K_t\mathcal{G})' + \mathcal{C}\mathcal{C}'.$$

- The last form is a sum of positive semidefinite matrices (numerically stable).

Appendix B.2: Likelihood, Moments, and Optimization

- **Prediction-error decomposition.** For $r < t$, a_r is a function of y^{t-1} and $E[a_t \mid y^{t-1}] = 0$, so $E[a_t a_r'] = 0$; being normal, innovations are independent. The map $y^T \mapsto a^T$ is triangular with unit Jacobian:

$$f(y_1, \ldots, y_T) = \prod_{t=1}^{T} (2\pi)^{-3/2} \det(\Omega_t)^{-1/2} \exp\Big(-\tfrac{1}{2} a_t' \Omega_t^{-1} a_t \Big).$$

- **Population moments.** $\Sigma = \mathcal{M}\Sigma\mathcal{M}' + \mathcal{C}\mathcal{C}'$; autocovariances $\Gamma_k = \mathcal{G}\mathcal{M}^k \Sigma \mathcal{G}'$.
- **Variance decomposition.** Shocks are independent, so $\Sigma = \sum_j \Sigma^{(j)}$ with $\Sigma^{(j)} = \mathcal{M}\Sigma^{(j)}\mathcal{M}' + \mathcal{C}_j\mathcal{C}_j'$; share of shock j in series i: $[\mathcal{G}\Sigma^{(j)}\mathcal{G}']_{ii} / [\mathcal{G}\Sigma\mathcal{G}']_{ii}$.
- **Reparametrization.** f_2, d, σ's $= \exp(\cdot)$; λ logistic; $a_2, D_w, D_N = \tanh(\cdot)$; $a_1 = (1 - a_2)\tanh(z)$ keeps each AR(2) stationary. If the Riccati equation fails, M is unstable, or some $\Omega_t \not\succ 0$: $\log L = -10^{10}$.

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